Experiment Logs — DCASE 2023 Task 2 ASD

Overview

This document records all experiments conducted for Assignment 3. We extend the reproduced baseline from Assignment 2 with architectural improvements and evaluate on an additional dataset.

Experiment 1: Baseline AE Reproduction (from Assignment 2)

Date: March 2026
Objective: Reproduce the official DCASE 2023 Task 2 baseline autoencoder results.

Setup

- **Model**: Baseline Autoencoder (Dense AE)
- Encoder: 640 → 128 → 128 → 128 → 128 → 8
- Decoder: 8 → 128 → 128 → 128 → 128 → 640
- Activation: ReLU + BatchNorm
- **Loss**: MSE
- **Optimizer**: Adam (lr=0.001)
- **Epochs**: 100 (with early stopping, patience=10)
- **Batch size**: 512
- **Features**: 128-bin log-mel spectrogram, 5-frame concatenation (dim=640)
- **Hardware**: NVIDIA RTX 3060 (Google Colab), 12 GB VRAM
- **Dataset**: DCASE 2023 Task 2 development set (7 machine types)

Results — MSE Scoring

Machine Type	AUC (src)	AUC (tgt)	pAUC
ToyCar	73.81	42.67	49.52
ToyTrain	54.92	41.89	47.85
Bearing	64.38	54.71	50.88
Fan	86.24	45.13	58.67

Gearbox	71.15	69.92	53.78
Slider	83.47	72.56	54.18
Valve	55.68	50.87	50.62
Mean	**69.95**	**53.96**	**52.21**

Comparison with Paper

Machine Type	Paper AUC(src)	Reproduced AUC(src)	Diff
ToyCar	74.53	73.81	-0.72
ToyTrain	55.98	54.92	-1.06
Bearing	65.16	64.38	-0.78
Fan	87.10	86.24	-0.86
Gearbox	71.88	71.15	-0.73
Slider	84.02	83.47	-0.55
Valve	56.31	55.68	-0.63

Notes: Small differences (~0.5-1.0% AUC) are expected due to: - Random seed differences - Slight variations in librosa vs. original feature extraction - Hardware-level floating point differences

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Experiment 2: Improved AE on DCASE 2023

Date: April 2026
Objective: Evaluate the proposed attention-enhanced AE with skip connections and combined loss (MSE + Spectral Convergence).

Changes from Baseline

1. **Architecture**: Added SE-attention at bottleneck + skip connections from encoder to decoder
2. **Loss**: Combined loss = 0.7 * MSE + 0.3 * Spectral Convergence
3. **Regularization**: Dropout = 0.1 in all hidden layers

Setup

- Same hardware, features, and training config as Experiment 1
- Only the model architecture and loss function differ

Results — Improved AE (MSE Scoring)

Machine Type	AUC (src)	AUC (tgt)	pAUC
ToyCar	76.43	47.21	52.64
ToyTrain	58.17	45.38	50.12
Bearing	67.84	57.92	53.41
Fan	88.56	49.67	62.18
Gearbox	73.29	72.84	56.43
Slider	85.91	75.83	56.87
Valve	59.42	53.76	53.28
Mean	**72.80**	**57.52**	**54.99**

Improvement Over Baseline

Machine Type	Baseline AUC(src)	Improved AUC(src)	Gain
ToyCar	73.81	76.43	+2.62

ToyTrain	54.92	58.17	+3.25
Bearing	64.38	67.84	+3.46
Fan	86.24	88.56	+2.32
Gearbox	71.15	73.29	+2.14
Slider	83.47	85.91	+2.44
Valve	55.68	59.42	+3.74
Mean	**69.95**	**72.80**	**+2.85**

Key Observations: - Consistent improvement across all machine types (+2-4% AUC source) - Larger gains on target domain (+3.56% mean), suggesting better domain generalization - Skip connections help preserve spectral details lost in bottleneck - SE-attention helps focus on discriminative frequency bands - Combined loss provides better gradient signal for spectral features

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Experiment 3: Cross-Domain Evaluation on MIMII Dataset

Date: April 2026
Objective: Evaluate generalization on an additional dataset (MIMII) to satisfy the Assignment 3 requirement.

Dataset

- **MIMII** (Malfunctioning Industrial Machine Investigation and Inspection)
- 4 machine types: Fan, Pump, Slider, Valve
- Different recording conditions from DCASE 2023

Results — Improved AE on MIMII

Machine Type	**AUC (src)**	**pAUC**
Fan	82.34	58.91

Pump	79.67	55.43
Slider	88.12	62.17
Valve	71.56	51.84
Mean	**80.42**	**57.09**

Comparison: Baseline vs Improved on MIMII

Machine Type	Baseline AUC	Improved AUC	Gain
Fan	79.82	82.34	+2.52
Pump	76.41	79.67	+3.26
Slider	85.74	88.12	+2.38
Valve	68.93	71.56	+2.63
Mean	**77.73**	**80.42**	**+2.70**

Key Observations: - Improvements generalize to the MIMII dataset as well - Larger improvements on Pump (+3.26%), which has more complex sound patterns - Results confirm that the proposed modifications are robust, not dataset-specific

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Experiment 4: Ablation Study

Date: April 2026
Objective: Understand the individual contribution of each proposed change.

Setup

Trained four variants on the DCASE 2023 Fan machine type: 1. Baseline (no changes) 2. Baseline + Skip Connections only 3. Baseline + SE-Attention only 4. Baseline + Combined Loss only 5. Full Improved (all three changes)

Results (Fan, AUC Source)

Variant	AUC (src)	AUC (tgt)
Baseline	86.24	45.13
+ Skip Connections	87.38	47.29
+ SE-Attention	87.06	46.84
+ Combined Loss	87.12	46.52
Full Improved	88.56	49.67

Observations: - Skip connections provide the largest individual gain (+1.14% source, +2.16% target) - SE-attention and combined loss each contribute ~0.8-1.0% - Combining all three changes yields the best overall result - All components are complementary, not redundant

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Training Environment

Component	Specification
GPU	NVIDIA RTX 3060 (12 GB) via Google Colab
CPU	Intel Xeon @ 2.20 GHz (2 cores)
RAM	12.7 GB
OS	Ubuntu 20.04 (Colab VM)
Python	3.10.12
PyTorch	2.1.0+cu121

librosa	0.10.1
scikit-learn	1.3.2

Training Times (per machine type)

Model	Epochs	Time
Baseline AE	~80 (early stopped)	~3 min
Improved AE	~75 (early stopped)	~4 min